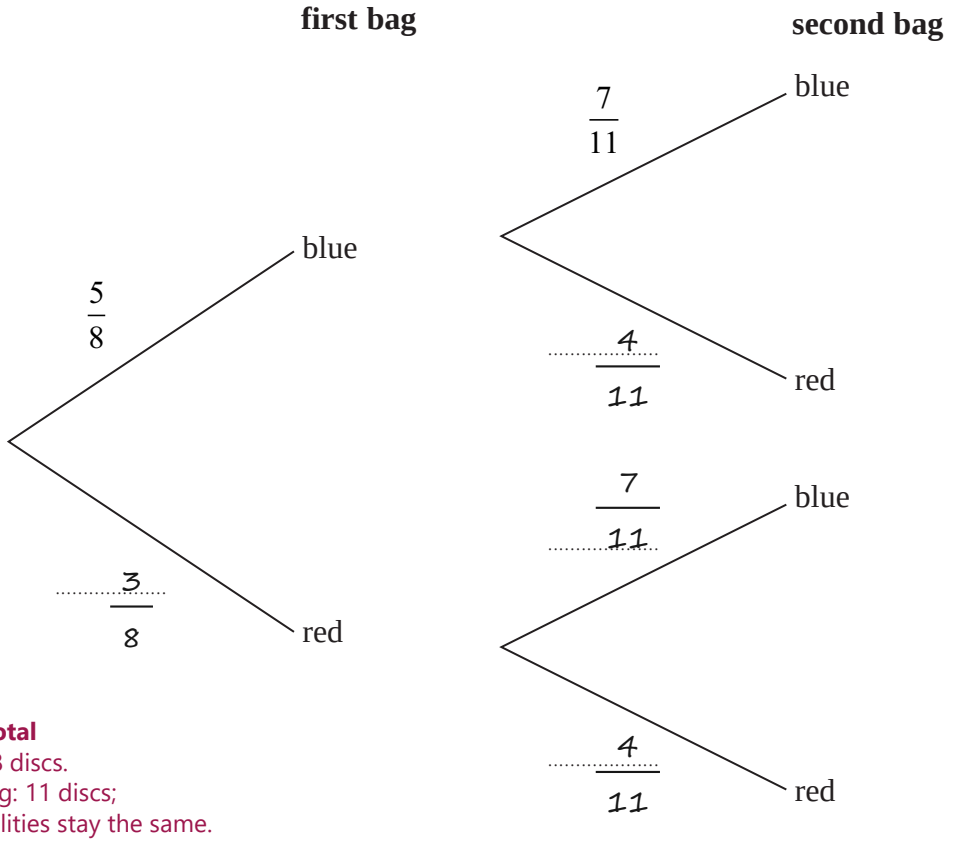
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**6** Kabita has two bags.

In the first bag there are 5 blue discs and 3 red discs.
In the second bag there are 7 blue discs and 4 red discs.

Kabita takes at random a disc from the first bag.
She then takes at random a disc from the second bag.

(a) Complete the probability tree diagram.




**2 marks total**
First bag: 8 discs.
Second bag: 11 discs;
its probabilities stay the same.

**(2)**

(b) Work out the probability that Kabita takes two red discs.

$$\frac{3}{8} \times \frac{4}{11} = \frac{12}{88} = \frac{3}{22}$$

**2 marks total**
Multiply along the red–red path.

$\frac{3}{22}$

**(2)**

**(Total for Question 6 is 4 marks)**